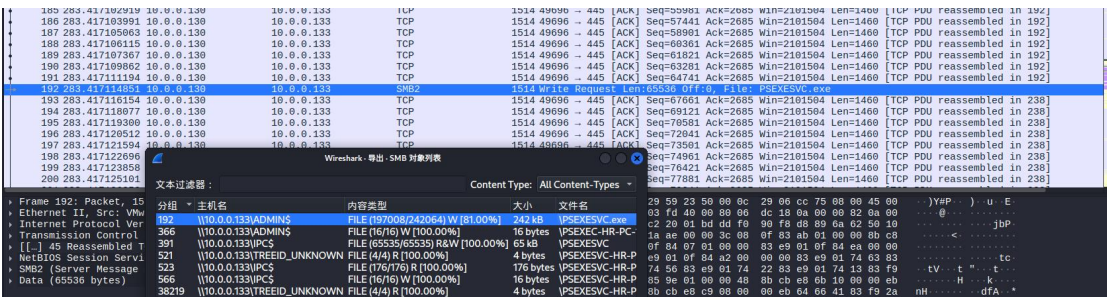


PsExec Hunt Lab

入侵检测系统（IDS）发出警报，指出存在涉及 PsExec 的可疑横向移动行为，表明可能存在未经授权的网络访问与数据传输。作为安全运营中心分析师，您的任务是分析提供的 PCAP 文件以追踪攻击者的活动轨迹：确定其入侵入口、目标设备、漏洞范围，并识别能揭示其在受感染环境中攻击策略与目标的关键特征。

1.为了有效追踪攻击者在我们网络中的活动，你能否识别攻击者最初获得访问权限的机器的 IP 地址？



No.	Time	Source	Destination	Protocol	Length	Info
185	283.417192919	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=709981 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 192]
186	283.417193891	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=74441 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 192]
187	283.417195063	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=589081 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 192]
188	283.417196115	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=60361 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 192]
189	283.417197367	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=61821 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 192]
190	283.417198602	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=63281 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 192]
191	283.417111194	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=64741 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 192]
192	283.417114051	10.0.0.130	10.0.0.133	SMB2	1514	Write Request Len=65536 Offset=0 File: \PSEXESVC.exe
193	283.417116154	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=67061 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]
194	283.417118077	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=69121 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]
195	283.417119300	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=70581 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]
196	283.417120512	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=72041 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]
197	283.417121594	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=73501 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]
198	283.417122696	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=74961 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]
199	283.417123858	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=76421 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]
200	283.417125101	10.0.0.130	10.0.0.133	TCP	1514	49690 → 445 [ACK] Seq=77881 Ack=2685 Win=2181504 Len=1460 [TCP PDU reassembled in 238]

Wireshark - 导出 - SMB 对象列表

Content Type: All Content-Types

帧 192: 数据包, 15 字节	以太网 II, Src: VMware	Internet Protocol Ver 4	Transmission Control	[...] 45 重新组装的 T	NetBIOS 会话服务	SMB2 (Server Message	Data (65536 bytes)
192	10.0.0.133\ADMIN\$	10.0.0.133\ADMIN\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$
366	10.0.0.133\ADMIN\$	10.0.0.133\ADMIN\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$
391	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$
521	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$
523	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$
566	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$
58219	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$	10.0.0.133\IPC\$

2.为了全面了解入侵的范围，你能确定攻击者最初转向的机器主机名吗？

过滤 ntlmssp.challenge.target_info

字符串搜分组详情 netbios computer name

ntlmssp.challenge.target_info						
分组详情		字符串		netbios computer name		
Options: 宽窄		区分大小写		Backwards Multiple occurrences		
No.	Time	Source	Destination	Protocol	Length	Info
131	283.408953905	10.0.0.133	10.0.0.130	SMB2	329	Session Setup Response
339	283.504443260	10.0.0.133	10.0.0.130	DCERPC	314	Bind_ack: call_id: 2,
534	284.520139857	10.0.0.133	10.0.0.130	SMB2	298	Session Setup Response
542	284.676472078	10.0.0.133	10.0.0.130	SMB2	298	Session Setup Response
550	284.754609339	10.0.0.133	10.0.0.130	SMB2	298	Session Setup Response
555	284.848040193	10.0.0.133	10.0.0.130	SMB2	298	Session Setup Response
38449	510.841851331	10.0.0.133	10.0.0.130	DCERPC	314	Bind_ack: call_id: 2,
38514	534.441660177	10.0.0.131	10.0.0.130	SMB2	369	Session Setup Response
38534	536.505360681	10.0.0.131	10.0.0.130	SMB2	369	Session Setup Response
38742	536.610057056	10.0.0.131	10.0.0.130	DCERPC	354	Bind_ack: call_id: 2,
39065	537.516289772	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Response
39070	537.829144599	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Response
39074	537.859416791	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Response
39091	546.004899370	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Response
39111	548.091599621	10.0.0.131	10.0.0.130	SMB2	369	Session Setup Response
39323	548.157945858	10.0.0.131	10.0.0.130	DCERPC	354	Bind_ack: call_id: 2,
39651	549.089868740	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Response


```

.... .0.. = Encrypt: False
Blob Offset: 0x00000048
Blob Length: 168
Security Blob [...]: 4e544c4d5353500002000000100010003800000015828ae27a41450c50b5aa3d000e
  NTLM Secure Service Provider
    NTLMSSP identifier: NTLMSSP
    NTLM Message Type: NTLMSSP_CHALLENGE (0x00000002)
      Target Name: SALES-PC
      Negotiate Flags: 0xe28a8215, Negotiate 56, Negotiate Key Exchange, Negotiate 128,
      NTLM Server Challenge: 7a41450c50b5aa3d
      Reserved: 0000000000000000
      Target Info
        Length: 96
        Maxlen: 96
        Offset: 72
        Attribute: NetBIOS domain name: SALES-PC
        Attribute: NetBIOS computer name: SALES-PC
        Attribute: DNS domain name: Sales-PC

```

3.知道攻击者用于认证的账户用户名，将帮助我们了解泄露的严重程度。攻击者用来认证的账户用户名是什么？

ntlmssp						
No.	Time	Source	Destination	Protocol	Length	Info
39679	552.523595599	10.0.0.130	10.0.0.131	SMB2	582	Session Setup Request, NTLMSSP_AUTH, User: \jdoe
39680	554.563272335	10.0.0.130	10.0.0.131	SMB2	228	Session Setup Request, NTLMSSP_NEGOTIATE
39700	554.563970067	10.0.0.130	10.0.0.131	SMB2	629	Session Setup Request, NTLMSSP_AUTH, User: .\IEUser
39905	554.626369898	10.0.0.130	10.0.0.131	DCERPC	218	Bind: call_id: 2, Fragment: Single, 2 context items: SVCCTL V2.0 (32bit NDR), SVCCTL V2.0 (6cb71c2c-8
39907	554.627762918	10.0.0.130	10.0.0.131	DCERPC	524	AUTH3: call_id: 2, Fragment: Single, NTLMSSP_AUTH, User: .\IEUser
40241	555.570239176	10.0.0.130	10.0.0.131	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE
40243	555.571305969	10.0.0.130	10.0.0.131	SMB2	261	Session Setup Request, NTLMSSP_AUTH, User: \
40245	555.572919474	10.0.0.130	10.0.0.131	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE
40247	555.573971849	10.0.0.130	10.0.0.131	SMB2	261	Session Setup Request, NTLMSSP_AUTH, User: \
40250	555.805200504	10.0.0.130	10.0.0.131	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE
40252	555.806347561	10.0.0.130	10.0.0.131	SMB2	261	Session Setup Request, NTLMSSP_AUTH, User: \
40255	555.850037084	10.0.0.130	10.0.0.131	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE
40257	555.850799050	10.0.0.130	10.0.0.131	SMB2	261	Session Setup Request, NTLMSSP_AUTH, User: \
130	283.488340907	10.0.0.130	10.0.0.133	SMB2	228	Session Setup Request, NTLMSSP_NEGOTIATE
132	283.488462212	10.0.0.130	10.0.0.133	SMB2	595	Session Setup Request, NTLMSSP_AUTH, User: \sales

4.在弄清攻击者如何在我们网络中移动后，我们需要知道他们在目标机器上做了什么。攻击者在目标上设置的服务执行文件叫什么名字？

Wireshark - 导出 - SMB 对象列表

文本过滤器: Content Type: All Content-Types

	内容类型	大小	文件名
0.133\ADMIN\$	FILE (197008/242064) W [81.00%]	242 kB	\PSEXESVC.exe
0.133\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-1C6C5D14.key
0.133\IPC\$	FILE (65535/65535) R&W [100.00%]	65 kB	\PSEXESVC
0.133\TREEID_UNKNOWN	FILE (4/4) R [100.00%]	4 bytes	\PSEXESVC-HR-PC-7980-stdout
0.133\IPC\$	FILE (176/176) R [100.00%]	176 bytes	\PSEXESVC-HR-PC-7980-stdout
0.133\IPC\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXESVC-HR-PC-7980-stdin
0.133\TREEID_UNKNOWN	FILE (4/4) R [100.00%]	4 bytes	\PSEXESVC-HR-PC-7980-stderr
0.133\IPC\$	FILE (128/128) R [100.00%]	128 bytes	\PSEXESVC-HR-PC-7980-stderr
0.131\ADMIN\$	FILE (242064/242064) W [100.00%]	242 kB	\PSEXESVC.exe
0.131\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-8FF87B23.key
0.131\IPC\$	FILE (65535/65535) R&W [100.00%]	65 kB	\PSEXESVC
0.131\ADMIN\$	FILE (242064/242064) W [100.00%]	242 kB	\PSEXESVC.exe
0.131\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-AF58F077.key
0.131\IPC\$	FILE (65535/65535) R&W [100.00%]	65 kB	\PSEXESVC
0.131\ADMIN\$	FILE (242064/242064) W [100.00%]	242 kB	\PSEXESVC.exe
0.131\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-CF174DD5.key

5.我们需要知道攻击者如何在被攻破的机器上安装该服务，以了解攻击者的横向移动策略。这有助于识别其他受影响的系统。PsExec 用哪个网络共享来在目标机器上安装该服务？

ADMIN\$ Windows 默认的管理共享

IPC\$ 进行 RPC 通信

40233	555.046960027	10.0.0.131	10.0.0.130	SMB2	158 GetInfo Response, File: PSEXESVC.exe
40234	555.047159611	10.0.0.130	10.0.0.131	SMB2	146 Close Request, File: PSEXESVC.exe
40235	555.047496942	10.0.0.131	10.0.0.130	SMB2	182 Close Response, File: PSEXESVC.exe
40236	555.050803304	10.0.0.130	10.0.0.131	TCP	60 49713 → 49669 [RST, ACK] Seq=236
40237	555.051200723	10.0.0.130	10.0.0.131	TCP	60 49712 → 135 [RST, ACK] Seq=329
40238	555.051309066	10.0.0.130	10.0.0.131	SMB2	146 Close Request, File: PSEXESVC-HR-PC-7980-stdout
40239	555.051554100	10.0.0.131	10.0.0.130	SMB2	182 Close Response, File: PSEXESVC-HR-PC-7980-stdout
40240	555.091991089	10.0.0.130	10.0.0.131	TCP	60 49711 → 445 [ACK] Seq=384414 Ack=445
40241	555.570239176	10.0.0.130	10.0.0.131	SMB2	186 Session Setup Request, NTLMSSP_A
40242	555.570867433	10.0.0.131	10.0.0.130	SMB2	338 Session Setup Response, Error: S
40243	555.571305969	10.0.0.130	10.0.0.131	SMB2	261 Session Setup Request, NTLMSSP_A
40244	555.571911512	10.0.0.131	10.0.0.130	SMB2	130 Session Setup Response, Error: S
40245	555.572919474	10.0.0.130	10.0.0.131	SMB2	186 Session Setup Request, NTLMSSP_A
40246	555.573404864	10.0.0.131	10.0.0.130	SMB2	338 Session Setup Response, Error: S
40247	555.573871848	10.0.0.131	10.0.0.131	SMB2	261 Session Setup Request, NTLMSSP_A

```

Channel Sequence: 0
Reserved: 0000
Command: Close (0)
Credits requested: 1
Flags: 0x00000030, Priority
Chain Offset: 0x00000000
Message ID: 58
Reserved: 0x0000feff
Tree Id: 0x00000001 \\10.0.0.131\ADMIN$
Session Id: 0x00002c0000000031 Acct:IEUser Domain:. Host:HR-PC
Signature: 00000000000000000000000000000000
[Response in: 40235]
[Time to response: 337.331 microseconds]
Close Request (0x06)
StructureSize: 0x0018
Close Flags: 0x0000
GUID handle, File: PSEXESVC.exe

```

6.我们必须确定用于两台机器之间通信的网络共享。PsExec 用哪个网络共享进行通信？


```
40238 555.051309066 10.0.0.130 10.0.0.131 SMB2 146 Close Request, File: PSEXESVC-HR-PC-3348-std
40239 555.051554100 10.0.0.131 10.0.0.130 SMB2 182 Close Response, File: PSEXESVC-HR-PC-3348-std
40240 555.091991089 10.0.0.130 10.0.0.131 TCP 60 49711 - 445 [ACK] Seq=384414 Ack=141779 Win=
40241 555.570239176 10.0.0.130 10.0.0.131 SMB2 186 Session Setup Request, NTLMSSP_NEGOTIATE
40242 555.570867433 10.0.0.131 10.0.0.130 SMB2 338 Session Setup Response, Error: STATUS_MORE_P
40243 555.571305969 10.0.0.130 10.0.0.131 SMB2 261 Session Setup Request, NTLMSSP_AUTH, User: \
40244 555.571911512 10.0.0.131 10.0.0.130 SMB2 130 Session Setup Response, Error: STATUS_ACCESS
40245 555.572919474 10.0.0.130 10.0.0.131 SMB2 186 Session Setup Request, NTLMSSP_NEGOTIATE
40246 555.573404864 10.0.0.131 10.0.0.130 SMB2 338 Session Setup Response, Error: STATUS_MORE_P
40247 555.573871848 10.0.0.130 10.0.0.131 SMB2 261 Session Setup Request, NTLMSSP_AUTH, User: \

Channel Sequence: 0
Reserved: 0000
Command: Close (0)
Credits requested: 1
Flags: 0x00000030, Priority
Chain Offset: 0x00000000
Message ID: 59
Reserved: 0x0000feff
Tree Id: 0x00000005 \\10.0.0.131\IPC$
Session Id: 0x00002c00000000031 Acct:IEUser Domain:. Host:HR-PC
Signature: 00000000000000000000000000000000
[Response in: 40238]
[Time to response: 245.034 microseconds]
Close Request (0x06)
StructureSize: 0x0018
Close Flags: 0x0000
GUID handle, File: PSEXESVC-HR-PC-3348-stdin
```

7.现在我们对攻击者在被攻破机器上的活动有了更清晰的了解，识别任何进一步的横向移动非常重要。攻击者在我们网络中针对的第二台机器的主机名是什么？

Wireshark - 导出: SMB 对象列表

文本过滤器: Content Type: All Content-Types

分组	主机名	内容类型	大小	文件名
192	\\10.0.0.133\ADMIN\$	FILE (197008/242064) W [81.00%]	242 kB	\PSEXESVC.exe
366	\\10.0.0.133\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-
391	\\10.0.0.133\IPC\$	FILE (65535/65535) R&W [100.00%]	65 kB	\PSEXESVC
521	\\10.0.0.133\TREEID_UNKNOWN	FILE (4/4) R [100.00%]	4 bytes	\PSEXESVC-HR-P
523	\\10.0.0.133\IPC\$	FILE (176/176) R [100.00%]	176 bytes	\PSEXESVC-HR-P
566	\\10.0.0.133\IPC\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXESVC-HR-P
38219	\\10.0.0.133\TREEID_UNKNOWN	FILE (4/4) R [100.00%]	4 bytes	\PSEXESVC-HR-P
38221	\\10.0.0.133\IPC\$	FILE (128/128) R [100.00%]	128 bytes	\PSEXESVC-HR-P
38593	\\10.0.0.131\ADMIN\$	FILE (242064/242064) W [100.00%]	242 kB	\PSEXESVC.exe
38767	\\10.0.0.131\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-
38794	\\10.0.0.131\IPC\$	FILE (65535/65535) R&W [100.00%]	65 kB	\PSEXESVC
39173	\\10.0.0.131\ADMIN\$	FILE (242064/242064) W [100.00%]	242 kB	\PSEXESVC.exe
39350	\\10.0.0.131\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-
39377	\\10.0.0.131\IPC\$	FILE (65535/65535) R&W [100.00%]	65 kB	\PSEXESVC
39756	\\10.0.0.131\ADMIN\$	FILE (242064/242064) W [100.00%]	242 kB	\PSEXESVC.exe
39934	\\10.0.0.131\ADMIN\$	FILE (16/16) W [100.00%]	16 bytes	\PSEXEC-HR-PC-
39961	\\10.0.0.131\IPC\$	FILE (65535/65535) R&W [100.00%]	65 kB	\PSEXESVC

一个 133，一个 131，第二台是 131

同 2 间

ntlmssp.challenge.target_info

分组详情

字符串

netbios computer name

Options: 宽窄

区分大小写

Backwards

Multiple occurrences

No.	Time	Source	Destination	Protocol	Length	Info
39065	537.516289772	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39070	537.829144599	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39074	537.859416791	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39091	546.004899370	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39111	548.091599621	10.0.0.131	10.0.0.130	SMB2	369	Session Setup Re
39323	548.157945858	10.0.0.131	10.0.0.130	DCERPC	354	Bind_ack: call_i
39651	549.089868740	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39655	549.092213165	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39660	549.355485643	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39664	549.386705964	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39678	552.523064724	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
39699	554.563582426	10.0.0.131	10.0.0.130	SMB2	369	Session Setup Re
39906	554.626786389	10.0.0.131	10.0.0.130	DCERPC	354	Bind_ack: call_i
40242	555.570867433	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
40246	555.573404864	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
40251	555.805728884	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re
40256	555.850380637	10.0.0.131	10.0.0.130	SMB2	338	Session Setup Re

NTLM Server Challenge: 185640357032d340
Reserved: 0000000000000000

Target Info

Length: 128
Maxlen: 128
Offset: 80

Attribute: NetBIOS domain name: MARKETING-PC
Target Info Item Type: NetBIOS domain name (0x0002)
Target Info Item Length: 24
NetBIOS Domain Name: MARKETING-PC

Attribute: NetBIOS computer name: MARKETING-PC
Target Info Item Type: NetBIOS computer name (0x0001)
Target Info Item Length: 24
NetBIOS Computer Name: MARKETING-PC